

Lean Manufacturing and the 7 Mudras

Waiting

Overproduction

Overprocessing

Defects

Motion

Inventory

Transportation

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Activities NOT related to adding value to the customer.

Value added activities are the those that the customer is paying for.

Non-value added activities are everything else.



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Waiting

A stoppage at one processing step because of a delay in a preceding processing step.

This can be caused by many things: Unplanned events, an imbalance between processing steps, poor planning, insufficient or unclear work instructions, etc.

Waiting extends lead times and lowers an organization's return on investment (ROI).



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Overproduction

Producing more than needed or earlier than needed.

This is commonly done to improve a localized goal like “machine efficiency” or “fixed cost per piece”.

Overproduction ties up equipment and people needed for more valuable tasks, and creates idle inventory.



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Overprocessing

The use of tolerances or appearance criteria, oversized or sophisticated equipment, or additional processing steps with costs in excess of what the customer has requested or is willing to pay for.

This is often caused by a lack of communication between the true customer and true supplier.

Overprocessing costs and lead time without benefit. It results in a loss of net earnings, sustainability and customer satisfaction.



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Defects

Products that do not conform to the customers' specifications or expectations.

Defects are caused by an inadequate process design or deviations from that design.

Defects result in a loss of the raw materials, labor, and utilities used to make the part, and extend the lead time while the replacement products are manufactured.



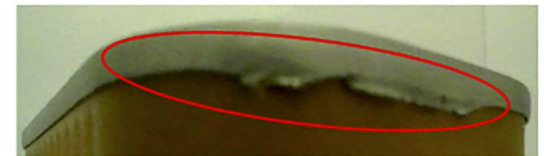
(a)



(b)



(c)



(d)



(e)



(f)

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Motion

All movement within a processing step that does not add value to the customer (e.g. only the last turn of a nut tightens the bolt.)

Wasted motion is caused by poor product or process design.

Wasted motion can lead to injuries, repetitive stress disorders, loss of morale, diminished earnings, and long lead times.



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Inventory

The costs associated with purchasing, storing, organizing, transporting and packaging unprocessed and unshipped inventory.

Excess inventory is often added to a process by design to mask other shortcomings in the system like defects, loss from damage, localized goals, etc.

Inventory waste adds costs and diminishes earnings.



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Transportation

Transportation within a process adds no value to the customer.

Wasted transportation is caused by poor process design or facility layout.

Wasted transportation adds costs and the risk of product damage.

